

Einstein field equations, the Schwarzschild solution and experimental tests of General Relativity

Important milestones in the early history of General Relativity were the Einstein field equations, Schwarzschild's solution to them and the observational consequences of this solution. The Schwarzschild solution¹ describes the space-time in the vicinity of a static, spherically symmetric mass, like the Sun, and the observational tests of this solution include the precession of the perihelion of planetary orbits – in particular the orbit of Mercury – and the bending of light in a gravitational field. A more recent test is the so-called radar echo delay of a signal sent from one planet (Earth) and reflected back from another one. An additional test of General Relativity, which depends only on the Equivalence Principle and not on the field equations, is the gravitational red-shift of light. The successful passing of these tests established General Relativity as the 'correct' theory of gravity. A feature of the Schwarzschild solution, not emphasised in the early days but given great prominence since, is the presence of the 'Schwarzschild' radius, which is the signature for the phenomenon of black holes. These matters are the concerns of this chapter. We begin with a comparison of the geodesic equation and the Newtonian limit of a weak, static gravitational field.

5.1 Newtonian limit

Consider the case of a weak, static field (such as, to a good approximation, that of the Sun) and a particle moving slowly in it ($v \ll c$). With $x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$, an *inertial* frame is one in which the metric tensor is

$$g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1),$$

so that $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$. A *weak* field is one for which

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad (5.1)$$

with $|h_{\mu\nu}| \ll 1$; each element of $g_{\mu\nu}$ is close to its inertial value. Non-relativistic motion, on the other hand, implies that $t \approx \tau$, $\frac{dx^0}{d\tau} \approx c$, $\frac{dx^i}{d\tau} \approx v^i \ll c$, so the geodesic Equation (4.18) with $\rho = i$ becomes

$$\frac{1}{c^2} \frac{d^2 x^i}{dt^2} + \Gamma^i_{00} = 0 \quad (5.2)$$

¹ Schwarzschild (1916a).

since the terms in $\frac{dx^j}{dt}$ are neglected. (We may write this equation as

$$\frac{d^2 x^j}{dt^2} = a^j = -c^2 \Gamma^j_{00};$$

then the right hand side represents the ‘gravitational force’, which gives the particle its acceleration. It is better, however, to discard the use of the term ‘force’ and accustom ourselves to thinking instead of gravity implying a curved space-time, in which particles move *freely*.) The connection coefficient in (5.2) is

$$\begin{aligned}\Gamma^j_{00} &= \frac{1}{2} g^{j\sigma} (2 g_{0\sigma,0} - g_{00,\sigma}) = -\frac{1}{2} g^{jk} g_{00,k} \\ &\approx -\frac{1}{2} \eta^{jk} \frac{\partial h_{00}}{\partial x^k} = -\frac{1}{2} \nabla_j h_{00},\end{aligned}\quad (5.3)$$

where the second and third equalities follow because the field is static, the fourth because we substitute (5.1), noting that η is constant, but keeping only first order terms in η , and the final equality since $\eta^{jk} = \delta^{jk}$ for spacelike indices. Putting (5.3) into (5.2) gives

$$\frac{d^2 x^j}{dt^2} = \frac{c^2}{2} \nabla_j h_{00}. \quad (5.4)$$

This is to be compared with Newton’s equation (see (1.1)–(1.4))

$$\frac{d^2 \mathbf{x}}{dt^2} = \mathbf{g} = -\nabla \phi \quad (5.5)$$

where ϕ is the gravitational potential. Comparison of (5.4) and (5.5) gives

$$h_{00} = -\frac{2\phi}{c^2}$$

and hence

$$g_{00} = -\left(1 + \frac{2\phi}{c^2}\right). \quad (5.6)$$

We have found one component of the metric tensor $g_{\mu\nu}$! Actually, this is all we can find by comparing Einstein’s theory with Newton’s, since Newton’s theory describes gravitation by means of one function only, the scalar field ϕ . (This naturally raises the question, how do we find the other components of $g_{\mu\nu}$?)

At a distance r from a gravitating body of mass M , we have $\phi = -\frac{MG}{r}$, so $g_{00} = -\left(1 - \frac{2GM}{rc^2}\right)$ and

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \dots \quad (5.7)$$

The ‘time-time’ component of the metric tensor has been found. This is actually sufficient to derive the gravitational red-shift, which is therefore a consequence of the Equivalence Principle alone, since the above reasoning relies only on this. We now turn our attention to the field equations, which will yield the other components of $g_{\mu\nu}$ (answering the question above).

5.2 Einstein field equations

The question to be answered is: given a particular matter distribution, what are the $g_{\mu\nu}$? There are two considerations to be borne in mind:

- (i) the equations we find must be tensor equations, and the tensors we have to hand, to describe the gravitational field, are $R^\kappa_{\lambda\mu\nu}$, $R_{\mu\nu}$ and R ;
- (ii) in a larger context we may summarise the present situation in Table 5.1.

The field equations we are looking for are the relativistic generalisations of Laplace's and Poisson's equations, in the cases of a vacuum and matter, respectively; and they should therefore yield these equations in the non-relativistic limit. Laplace's and Poisson's equations are second order differential equations, and we have seen that in the weak field limit ϕ becomes essentially g_{00} , so the relativistic field equations should be second order differential equations in $g_{\mu\nu}$; and, indeed, the tensors above, $R^\kappa_{\lambda\mu\nu}$, $R_{\mu\nu}$ and R , all contain second order derivatives of $g_{\mu\nu}$, since $R^\kappa_{\lambda\mu\nu}$ involves derivatives of $\Gamma^\kappa_{\lambda\mu}$ and $\Gamma^\kappa_{\lambda\mu}$ involves derivatives of $g_{\mu\nu}$. Apart from these guidelines, finding the field equations is a matter of guess-work. Let us begin with the vacuum field equations.

5.2.1 Vacuum field equations

The right hand side will clearly be zero, since there is no matter. The equation $R^\kappa_{\lambda\mu\nu} = 0$ is no good, though, because that would imply that outside a massive body space-time is flat and there is no gravitational field. So let us consider the equations

$$R_{\mu\nu} = 0. \quad (5.8)$$

These are 10 equations for 10 unknowns $g_{\mu\nu}$. They do *not* imply that $R^\kappa_{\lambda\mu\nu} = 0$, since as we saw in Chapter 4 the Riemann tensor has 20 independent components whereas the Ricci tensor only has 10, so that Equations (5.8) would constitute 10 relations among the 20 independent components of the Riemann tensor. The Equations (5.8) are in fact the vacuum Einstein field equations. Their general solution is unknown, but particular solutions are known, corresponding to cases with particular symmetry, for example the Schwarzschild

Table 5.1

	Non-relativistic	Relativistic
Equations of motion	Newton: $\frac{d^2 x^i}{dt^2} = -\nabla_i \phi$	Geodesic equation: $\frac{d^2 x^\mu}{ds^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{ds} \frac{dx^\rho}{ds} = 0$
Field equations	Laplace: $\nabla^2 \phi = 0$ vacuum Poisson: $\nabla^2 \phi = 4\pi\rho G$ matter	Einstein field equations

solution. We shall show below that the non-relativistic limit of (5.8) gives Laplace's equation.

As far as the matter field equations are concerned, it is clear that the right hand side must be a tensor describing the matter distribution, which in the non-relativistic limit reduces essentially to ρ , as in Poisson's equation. This tensor is called the energy-momentum tensor.

5.2.2 Energy-momentum tensor

The energy-momentum tensor (or stress-energy-momentum tensor, or stress-energy tensor), denoted $T_{\mu\nu}$, is a rank 2 tensor. It is not to be confused with the energy-momentum 4-vector (a rank 1 tensor). In this section we exhibit $T_{\mu\nu}$ for *dust*, that is, incoherent matter whose particles do not interact. The definition is

$$T^{\mu\nu} = \rho_0 u^\mu u^\nu \quad (5.9)$$

or more properly

$$T^{\mu\nu}(x) = \rho_0(x) u^\mu(x) u^\nu(x), \quad (5.10)$$

where ρ_0 is the proper density of matter, that is the density moving with the flow, and u^μ is its 4-velocity

$$u^\mu = \frac{1}{c} \frac{dx^\mu}{d\tau}. \quad (5.11)$$

We then have

$$ds^2 = -c^2 d\tau^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 = -c^2 dt^2 \left(1 - \frac{v^2}{c^2}\right)$$

and hence

$$\frac{d\tau}{dt} = \left(1 - \frac{v^2}{c^2}\right)^{1/2} = \frac{1}{\gamma},$$

so that

$$T^{00} = \rho_0 \left(\frac{dt}{d\tau}\right)^2 = \gamma^2 \rho_0 = \rho. \quad (5.12)$$

Here ρ is the mass density in a moving frame. The two factors of γ are easy to understand: in Special Relativity the mass (strictly, mass-energy) of moving material increases over its rest value by a factor γ , and the volume decreases by the same factor. So $c^2 T^{00}$ is the relativistic energy density of matter. Similarly

$$T^{0i} = \rho_0 u^0 u^i = \rho_0 \frac{1}{c^2} \frac{dx^0}{d\tau} \frac{dx^i}{d\tau} = \gamma^2 \rho_0 \frac{v^i}{c} = \rho \frac{v^i}{c}, \quad (5.13)$$

where $v^i = \frac{dx^i}{dt}$. And finally

$$T^{ik} = \rho_0 \frac{1}{c^2} \frac{dx^i}{d\tau} \frac{dx^k}{d\tau} = \gamma^2 \rho_0 \frac{v^i v^k}{c^2} = \rho \frac{v^i v^k}{c^2}. \quad (5.14)$$

Putting these components together we may display $T^{\mu\nu}$ as a matrix

$$T^{\mu\nu} = \rho \begin{pmatrix} 1 & \frac{v_x}{c} & \frac{v_y}{c} & \frac{v_z}{c} \\ \frac{v_x}{c} & \frac{v_x^2}{c^2} & \frac{v_x v_y}{c^2} & \frac{v_x v_z}{c^2} \\ \frac{v_y}{c} & \frac{v_y v_x}{c^2} & \frac{v_y^2}{c^2} & \frac{v_y v_z}{c^2} \\ \frac{v_z}{c} & \frac{v_z v_x}{c^2} & \frac{v_z v_y}{c^2} & \frac{v_z^2}{c^2} \end{pmatrix}. \quad (5.15)$$

In the non-relativistic approximation the dominant component of $T^{\mu\nu}$ is $T^{00} \approx \rho \approx \rho_0$, the density of matter in space.

Energy and momentum are of course conserved quantities, so we expect $T^{\mu\nu}$ to obey a conservation law. We need to find the form of this law, whatever it is, in General Relativity. In Special Relativity it is

$$T^{\mu\nu}{}_{;\nu} = 0. \quad (5.16)$$

To see that this is indeed a conservation law, put $\mu = 0$:

$$\begin{aligned} T^{00}{}_{;0} + T^{0j}{}_{;j} &= 0, \\ \frac{1}{c} \frac{\partial \rho}{\partial t} + \frac{1}{c} \frac{\partial}{\partial x^j} (\rho v^j) &= 0, \\ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) &= 0. \end{aligned} \quad (5.17)$$

This expresses the law of conservation of energy (strictly mass-energy). The reader might find it useful to be reminded of a similar law in electrodynamics, that of the conservation of charge Q . Suppose the charge inside a bounded volume V is Q , then if ρ is the *charge* density (see Fig 5.1),

$$Q = \int \rho \, dV.$$

In any particular frame there are also moving charges so there is also a current density

$$\mathbf{j} = \rho \mathbf{v}$$

where $\mathbf{v}$ is the velocity of the moving charge. Charge will then flow out of (or into) the bounding volume V through its surface S , so that the total charge enclosed in V will in general change. In a time δt the change in the charge enclosed is

$$\delta Q_1 = \frac{\partial Q}{\partial t} \delta t = \left\{ \int \frac{\partial \rho}{\partial t} \, dV \right\} \delta t.$$

The charge escaping through the surface is, on the other hand,

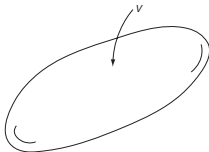


Fig. 5.1

A volume V containing a total charge Q , with charge density ρ .

$$\delta Q_2 = \left\{ \int \mathbf{j} \cdot d\mathbf{S} \right\} \delta t = \left\{ \int \mathbf{v} \cdot \mathbf{j} dV \right\} \delta t,$$

where the first integral is taken over the closed surface and the second equality follows from Gauss's theorem. The law of charge conservation is $\delta Q_1 + \delta Q_2 = 0$, i.e.

$$\int \left(\frac{\partial \rho}{\partial t} + \mathbf{v} \cdot \mathbf{j} \right) dV = 0,$$

which implies that, in differential form,

$$\frac{\partial \rho}{\partial t} + \mathbf{v} \cdot (\rho \mathbf{v}) = 0$$

where ρ is the *charge* density and $\mathbf{v}$ its velocity at any point. This equation is of the same form as (5.17) above, which therefore expresses the conservation of energy, or more strictly, mass-energy.

Now put $\mu = i$ in Equation (5.16). This gives, successively,

$$\begin{aligned} T^{i0}{}_{,0} + T^{ik}{}_{,k} &= 0, \\ \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{1}{c} \rho v^i \right) + \frac{1}{c^2} \frac{\partial}{\partial x^k} (\rho v^i v^k) &= 0, \\ \frac{\partial}{\partial t} (\rho v^i) + \mathbf{v} \cdot (\rho \mathbf{v} v^i) &= 0. \end{aligned} \quad (5.18)$$

This is of the same form as (5.17) above, except that ρ , the energy density, has been replaced by ρv^i , the momentum density, so (5.18) expresses conservation of momentum (density). We have therefore shown that (5.16) expresses conservation of energy-momentum in Special Relativity. In General Relativity the conservation law follows by replacing the ordinary derivative by a covariant one:

$$T^{\mu\nu}{}_{;\nu} = 0. \quad (5.19)$$

Recall once more that the form of the energy-momentum tensor we have been considering is appropriate only for dust particles, on which no forces act: the particles do not interact.

A more general stress tensor describes a fluid, in which there is pressure (exerting a force and therefore an acceleration). There is also a stress tensor for the electromagnetic field, which is important in cosmology. This will be considered later.

5.2.3 Matter field equations

We claim that the relativistic version of Laplace's (vacuum) equation is $R_{\mu\nu} = 0$, Equation (4.8). We want the relativistic version of Poisson's equation, and it is clear that ρ should be replaced by $T^{\mu\nu}$. We might guess that the desired equation is

$$R_{\mu\nu} = \frac{8\pi G}{c^2} T_{\mu\nu} \quad (?) \quad (5.20)$$

Both sides of the equation are rank $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ tensors. (It is obvious that having defined $R_{\mu\nu}$ and $T_{\mu\nu}$, we can raise and lower indices at will. The only requirement is that in a given equation the tensorial type is the same throughout.) The constant $\frac{8\pi G}{c^2}$ will be justified later. Is the above equation credible? It is not, because $T^{\mu\nu}_{;\nu} = 0$ but $R^{\mu\nu}_{;\nu} \neq 0$. This latter fact follows from the Bianchi identities (4.73):

$$R^{\mu}_{\nu\rho\sigma;\lambda} + R^{\mu}_{\nu\sigma\lambda;\rho} + R^{\mu}_{\nu\lambda\rho;\sigma} = 0.$$

Putting $\mu = \rho$ and contracting gives

$$R_{\nu\sigma;\lambda} + R^{\mu}_{\nu\sigma\lambda;\mu} - R_{\nu\lambda;\sigma} = 0,$$

where the minus sign in the last term follows from the antisymmetry under $\lambda \leftrightarrow \rho$. On multiplying by $g^{\nu\sigma}$ we find, successively

$$\begin{aligned} R_{;\lambda} - R^{\mu}_{\lambda;\mu} - R^{\sigma}_{\lambda;\sigma} &= 0, \\ \delta^{\rho}_{\lambda} R_{;\rho} - 2 R^{\rho}_{\lambda;\rho} &= 0, \\ (\delta^{\rho}_{\lambda} R - 2 R^{\rho}_{\lambda})_{;\rho} &= 0, \\ (R^{\rho\lambda} - 1/2 g^{\rho\lambda} R)_{;\rho} &= 0; \end{aligned} \quad (5.21)$$

or

$$G^{\mu\nu}_{;\nu} = 0, \quad (5.22)$$

where

$$G^{\mu\nu} = R^{\mu\nu} - 1/2 g^{\mu\nu} R \quad (5.23)$$

is called the *Einstein tensor*. It is not the Ricci tensor, which has vanishing covariant divergence, but the Einstein tensor. Then, instead of (5.20) we propose

$$R_{\mu\nu} - 1/2 g_{\mu\nu} R = \frac{8\pi G}{c^2} T_{\mu\nu} \quad (5.24)$$



Bridge over the river Tisza in Szeged. Hungarian graffiti artists are clearly well educated but seem to use unusual units for the gravitational constant. I am grateful to Dr Geretovszkyné Varjú Katalin for sending this picture.

for the relativistic matter field equations. Note that in the *absence* of matter, $T_{\mu\nu} = 0$, (5.24) gives

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0,$$

which on multiplying by $g^{\mu\nu}$ gives $R - 2R = 0$, (since $g^{\mu\nu} g_{\mu\nu} = 4$), hence $R = 0$, and

$$R_{\mu\nu} = 0,$$

which are indeed the vacuum field equations (5.8).

We now have to show that the weak field non-relativistic limit of (5.24) gives Poisson's equation. First multiply (5.24) by $g^{\mu\nu}$:

$$R - 2R = \frac{8\pi G}{c^2} T \Rightarrow R = -\frac{8\pi G}{c^2} T$$

where $T = g^{\mu\nu} T_{\mu\nu}$. Hence the field equations (5.24) may be rewritten as

$$R_{\mu\nu} = \frac{8\pi G}{c^2} (T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T). \quad (5.25)$$

Neglecting terms in $\frac{v^2}{c^2}$ and $\rho \frac{v}{c}$, the energy momentum tensor becomes, from (5.15),

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

In the weak field approximation (5.1) above $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. To lowest order in h this implies that

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}, \quad (5.26)$$

since then

$$g^{\mu\kappa} g_{\kappa\nu} = (\eta^{\mu\kappa} - h^{\mu\kappa})(\eta_{\kappa\nu} + h_{\kappa\nu}) = \delta^{\mu}_{\nu} - h^{\mu}_{\nu} + h^{\mu}_{\nu} + O(h^2) = \delta^{\mu}_{\nu}$$

to lowest order (*linear approximation*), as required. Then, also to lowest order $T = -\rho$ and

$$T^{\mu\nu} - 1/2 g^{\mu\nu} T = \frac{\rho}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \frac{\rho}{2} \delta^{\mu\nu}; \quad (5.27)$$

and also

$$T_{\mu\nu} - 1/2 g_{\mu\nu} T \approx \eta_{\mu\rho} \eta_{\nu\kappa} (T^{\rho\kappa} - 1/2 T g^{\rho\kappa}) = \frac{\rho}{2} \delta_{\mu\nu}. \quad (5.28)$$

Now let us calculate $R_{\mu\nu}$. From (4.31) and (4.35) we have

$$R_{\mu\nu} = \Gamma^{\kappa}_{\mu\nu,\kappa} - \Gamma^{\kappa}_{\mu\kappa,\nu} + \Gamma^{\kappa}_{\rho\kappa} \Gamma^{\rho}_{\mu\nu} - \Gamma^{\kappa}_{\rho\nu} \Gamma^{\rho}_{\mu\kappa}. \quad (5.29)$$

To lowest (first) order in h the connection coefficients are

$$\Gamma^{\kappa}_{\mu\nu} = 1/2 g^{\kappa\sigma} (g_{\sigma\mu,\nu} + g_{\sigma\nu,\mu} - g_{\mu\nu,\sigma}) = 1/2 \eta^{\kappa\sigma} (h_{\sigma\mu,\nu} + h_{\sigma\nu,\mu} - h_{\mu\nu,\sigma})$$

so the third and fourth terms on the right hand side of (5.29) are of order h^2 and can be ignored in the linear approximation. We then have, as may easily be verified,

$$R_{\mu\nu} = 1/2 \eta^{\kappa\sigma} (h_{\sigma\nu,\mu\kappa} + h_{\mu\kappa,\sigma\nu} - h_{\mu\nu,\sigma\kappa} - h_{\sigma\kappa,\mu\nu}).$$

Thus, in the static approximation

$$\begin{aligned} R_{00} &= 1/2 \eta^{\kappa\sigma} (h_{\sigma 0,\mu 0} + h_{0\kappa,\sigma 0} - h_{00,\sigma\kappa} - h_{\sigma\kappa,00}) \\ &= -1/2 \eta^{\kappa\sigma} h_{00,\sigma\kappa} = -1/2 \left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 \right) h_{00} \\ &= -1/2 \nabla^2 h_{00}. \end{aligned}$$

From above $h_{00} = -\frac{2\phi}{c^2}$ and the above equation gives $R_{00} = \frac{1}{c^2} \nabla^2 \phi$. The field equation (5.25) with $\mu = \nu = 0$ then gives (see (5.28))

$$\frac{1}{c^2} \nabla^2 \phi = \frac{8\pi G}{c^2} \frac{\rho}{2} \Rightarrow \nabla^2 \phi = 4\pi G \rho,$$

which is Poisson's equation (1.6).

Before leaving this section it might be useful to reflect once more on Table 5.1, showing the equations of motion for test particles, and the gravitational field equations. It is clear that given a particular distribution of (heavy) matter, the curvature of space-time (that is, essentially, $g_{\mu\nu}$) is fixed by the field equations; and therefore the motion of test bodies is also fixed – they simply follow a geodesic in the curved space. Wheeler summarises this in one of his famous aphorisms: ‘matter tells space how to curve, and space tells matter how to move’. This might sound circular, but it isn’t because in the first statement the amount of matter involved is large – large enough to cause significant curvature in space-time, whereas the ‘matter’ in the second phrase is simply a test body, which is small enough not to give rise to any gravitational field of its own. It will be appreciated, however, that when the mass of a test body becomes large enough to have an effect on the spatial geometry, then the problem becomes complicated and non-linear. In fact the gravitational field equations themselves are non-linear and the only method of procedure is one based on an approximation scheme – a ‘post-Newtonian’ approximation (General Relativity being a post-Newtonian theory).

Einstein, Infeld and Hoffmann did a significant amount of work on this problem, which has become known as the *problem of motion*. They came to the interesting conclusion that the equations of motion in General Relativity were actually consequences of the field equations themselves; a situation which, for example, does not hold in electrodynamics. This general topic of investigation was of particular significance to Einstein since he took the view that a point mass was a singularity in the gravitational field, and this would link very intimately questions about the field and its equations with those of how a point mass moves in the field. From the point of view of modern particle physics, however, this point of view would seem too simple, since it appears that there are no massive particles in nature which do not *also* carry other field quantities. The electron carries charge, the neutrino weak isospin and weak hypercharge, the quarks colour; so these particles, on Einstein’s view, are also singularities in the electromagnetic, weak isospin and weak hypercharge, and colour fields. Perseverance with Einstein’s question would take us into the territory of unified field theories. These theories belong, however, to the quantum regime, whereas General Relativity is, at the level we are at present concerned, a classical theory. Einstein’s question then enters difficult and unknown territory.

5.3 Schwarzschild solution

We are concerned with a solution of the vacuum field equations $R_{\mu\nu}=0$, in the case corresponding to the Solar System; that is, to a very good approximation, the field produced by a static spherically symmetric body at rest. The static condition means that $g_{\mu\nu}$ is independent of x^0 ; and in addition ds^2 is invariant under $x^0 \rightarrow -x^0$ (time reversal), so there must be no terms involving $dx^i dx^0$ in the expansion of ds^2 . This means that $g_{i0}=g_{0i}=0$. With these conditions, the most general form of the space-time line element compatible with spherical symmetry is

$$ds^2 = -U(r) c^2 dt^2 + V(r) dr^2 + W(r) r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (5.30)$$

By virtue of the special symmetry conditions obtaining here, the ten components of the metric tensor, in general each dependent on all the x^μ , reduce to only three functions, and these are functions of r alone. The field equations are second order differential equations for these functions. In fact the three functions may be reduced to two. Recall that r is only a radial *parameter*, not an actual distance, so it may be replaced by any function of r . Put $Wr^2 = \hat{r}^2$. Then $\hat{r} = \sqrt{W}r$ and $\frac{d\hat{r}}{dr} = \sqrt{W} \left(1 + \frac{r}{2W} \frac{dW}{dr} \right)$, so

$$V dr^2 = \frac{V}{W} \left(1 + \frac{r}{2W} \frac{dW}{dr} \right)^{-2} d\hat{r}^2 \equiv \hat{V} d\hat{r}^2.$$

We may write, at the same time, $U(r) = \hat{U}(\hat{r})$. The effect of all this is to replace r by $\hat{r}$, to replace U and V by corresponding functions with hats, but W is replaced by unity in (5.30). Then removing the hats, and putting, at the same time,

$$U(r) = e^{2\nu(r)}, \quad V(r) = e^{2\lambda(r)},$$

the line element becomes

$$ds^2 = -e^{2\nu} c^2 dt^2 + e^{2\lambda} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (5.31)$$

The metric tensor, in covariant and contravariant versions, is then

$$g_{\mu\nu} = \begin{pmatrix} -e^{2\nu} & 0 & 0 & 0 \\ 0 & e^{2\lambda} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} -e^{-2\nu} & 0 & 0 & 0 \\ 0 & e^{-2\lambda} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}. \quad (5.32)$$

The two unknown functions $\nu(r)$ and $\lambda(r)$ will now be found from the vacuum field equations. We first find the connection coefficients: for example

$$\begin{aligned} \Gamma^1_{00} &= \frac{1}{2} g^{1\sigma} (2g_{0\sigma,0} - g_{00,\sigma}) = -\frac{1}{2} g^{11} g_{00,1} = -\frac{1}{2} e^{-2\lambda} \frac{\partial}{\partial r} (-e^{2\nu}) \\ &= \nu' e^{2\nu-2\lambda}, \end{aligned} \quad (5.33)$$

where $\nu' = \frac{d\nu}{dr}$. Similarly the other connection coefficients turn out to be

$$\begin{aligned} \Gamma^0_{10} &= \Gamma^0_{01} = \nu', \\ \Gamma^1_{11} &= \lambda', \quad \Gamma^1_{22} = -r e^{-2\lambda}, \quad \Gamma^1_{33} = -r \sin^2 \theta e^{-2\lambda}, \\ \Gamma^2_{12} &= \Gamma^2_{21} = \Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{r}, \quad \Gamma^2_{33} = -\sin \theta \cos \theta, \\ \Gamma^3_{23} &= \Gamma^3_{32} = \cot \theta, \quad \text{others} = 0. \end{aligned} \quad (5.34)$$

These are now to be substituted in the vacuum equations

$$R_{\mu\nu} = \Gamma^{\kappa}_{\mu\nu,\kappa} - \Gamma^{\kappa}_{\mu\kappa,\nu} + \Gamma^{\kappa}_{\rho\kappa} \Gamma^{\rho}_{\mu\nu} - \Gamma^{\kappa}_{\rho\nu} \Gamma^{\rho}_{\mu\kappa} = 0.$$

We have, consulting (5.33), (5.34) and remembering the static condition,

$$\begin{aligned} R_{00} &= \Gamma^{\kappa}_{00,\kappa} + \Gamma^{\kappa}_{0\kappa,0} + \Gamma^{\kappa}_{\rho\kappa} \Gamma^{\rho}_{00} - \Gamma^{\kappa}_{\rho 0} \Gamma^{\rho}_{0\kappa} \\ &= \Gamma^1_{00,1} + \Gamma^{\kappa}_{1\kappa} \Gamma^1_{00} - (\Gamma^1_{00} \Gamma^0_{01} + \Gamma^0_{10} \Gamma^1_{00}) \\ &= \frac{\partial}{\partial r} (v' e^{2v-2\lambda}) + v' e^{2v-2\lambda} \left(v' + \lambda' + \frac{2}{r} \right) - 2(v')^2 e^{2v-2\lambda} \\ &= e^{2v-2\lambda} \left(v'' + v'^2 - v'\lambda' + \frac{2v'}{r} \right). \end{aligned}$$

Then the field equations give

$$R_{00} = \left(v'' + v'^2 - v'\lambda' + \frac{2v'}{r} \right) e^{2v-2\lambda} = 0. \quad (\text{A})$$

Similarly,

$$R_{11} = -v'' + v'\lambda' + \frac{2\lambda'}{r} - v'^2 = 0, \quad (\text{B})$$

$$R_{22} = (-1 - r v' + r \lambda') e^{-2\lambda} + 1 = 0, \quad (\text{C})$$

$$\begin{aligned} R_{33} &= R_{22} \sin^2 \theta, \\ R_{\mu\nu} &= 0 \quad (\mu \neq \nu). \end{aligned} \quad (5.35)$$

These are three independent equations for the functions $v(r)$ and $\lambda(r)$. The factor $e^{2v-2\lambda}$ in Equation (A) is non-zero, so Equations (A) and (B) give, on adding

$$\lambda' + v' = 0 \Rightarrow \lambda(r) + v(r) = \text{const.}$$

As $r \rightarrow \infty$, however, the metric (5.32) must approach the Minkowski metric $\eta_{\mu\nu}$, so $\lambda, v \rightarrow 0$, hence the constant above is zero and

$$\lambda(r) = -v(r).$$

Then Equation (C) gives

$$(1 + 2rv')e^{2v} = 1 \Rightarrow (r e^{2v})' = 1 \Rightarrow r e^{2v} = r - 2m,$$

where $2m$ is a constant of integration. That is,

$$e^{2v} = 1 - \frac{2m}{r},$$

which on comparison with (5.32) gives

$$g_{00} = -\left(1 - \frac{2m}{r}\right), \quad g_{11} = \frac{1}{1 - \frac{2m}{r}}.$$

In the weak field approximation, however, $g_{00} = -\left(1 - \frac{2GM}{rc^2}\right)$, so we can identify the constant of integration as $m = \frac{GM}{c^2}$ and finally

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right)c^2 dt^2 + \frac{dr^2}{1 - \frac{2GM}{rc^2}} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (5.36)$$

or, equivalently,

$$ds^2 = -\left(1 - \frac{2m}{r}\right)c^2 dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2),$$

with

$$m = \frac{MG}{c^2}. \quad (5.37)$$

This is the Schwarzschild solution. It is *exact* (and, as a consequence, the weak field approximation found above is now seen also to be exact – an exact solution of the field equations). As $r \rightarrow \infty$ Schwarzschild space-time approaches Minkowski space-time, as desired. The solution holds for the space-time *outside* a body of mass M . We shall see that it gives small corrections to the Newtonian predictions for the motions of light and the planets. It is worth noting that, for this vacuum solution, M is simply the total mass of the ‘gravitating’ body; the actual *distribution* of matter inside the body is irrelevant. This is a feature shared with Newtonian physics, where for example the gravitational potential due to a mass M depends only on the mass, not on the distribution of matter inside it.

5.3.1 Apparent ‘singularity’ at $r = 2m$

A glance at (5.36) above shows that g_{11} , the coefficient of dr^2 , becomes singular as $r \rightarrow \frac{2GM}{c^2}$. What are the consequences of this? Does it give trouble? We shall see that the consequences are very far-reaching, giving rise to black holes, which have to be investigated carefully. Here, however, we may note that in the Solar System – which is what is relevant for many of the tests of General Relativity – this ‘singularity’ gives no trouble. Let me make the following observations:

- (i) For the Sun, $M = 1.99 \times 10^{30}$ kg, so

$$\frac{2GM}{c^2} = \frac{2 \times 1.99 \times 10^{30} \times 6.67 \times 10^{-11}}{9 \times 10^{16}} = 2.95 \text{ km}.$$

This quantity is called the Schwarzschild radius r_S , so for the Sun

$$r_S = \frac{2GM}{c^2} = 2m = 2.95 \text{ km}; \quad (5.38)$$

and we may write the Schwarzschild metric as

$$ds^2 = -\left(1 - \frac{r_S}{r}\right)c^2 dt^2 + \frac{dr^2}{1 - \frac{r_S}{r}} + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (5.39)$$

The question then is: do we have trouble when $r \rightarrow r_S$? The answer is No, since the Schwarzschild surface (a spherical surface) $r = r_S$ is *inside* the Sun, whose radius is $R = 6.96 \times 10^5$ km. So at $r = r_S$ the Schwarzschild solution above does *not hold*, since it is a solution to the *vacuum* field equations. The region in which General Relativity will be tested in the Solar System is that for which $r > R$, hence $\frac{r_S}{r} < 4.2 \times 10^{-6}$, and Schwarzschild space-time differs only very slightly from Minkowski space-time.

- (ii) We now know, however, that when stars become sufficiently old they collapse, and very heavy stars may collapse to $r < r_S$, a radius smaller than their Schwarzschild radius. In that case the surface $r = r_S$ is in the ‘vacuum’, outside the star, and the problems related to the Schwarzschild surface become real, and will be dealt with in Chapter 7, where we shall see that light cannot escape from a star whose radius is less than $2GM/c^2$. We may note here, however, that the ‘singularity’ $r = 2m$ is more like a coordinate singularity than an actual singularity in the geometry. By analogy, at the north and south poles on a sphere $g_{22} = \sin^2\theta$, $g^{22} = \frac{1}{\sin^2\theta} \rightarrow \infty$ as $\theta \rightarrow 0, \pi$. The (contravariant components of the) metric tensor may become singular, but nothing unusual appears at the poles; a sphere is, after all, a homogeneous space, and no one point is different from any other point.
- (iii) It is interesting to note that the Schwarzschild radius first appeared in the work of Laplace in 1798 (the fourth year of the French Republic). Consider a particle of mass m escaping from the gravitational pull of a body (planet, star) of mass M and radius R . Using a Newtonian argument, the work done to escape is

$$\int_R^\infty F dr = mMG \int_R^\infty \frac{dr}{r^2} = \frac{mMG}{R}.$$

Supposing light to consist of particles of mass m travelling at speed $v = c$ with kinetic energy $\frac{1}{2} mc^2$ (!), we may then deduce that the light will not escape if $\frac{mMG}{R} > \frac{1}{2} mc^2$,

i.e. if $R < \frac{2GM}{c^2}$, precisely the Schwarzschild radius! If the radius of a star is smaller than its Schwarzschild radius, it will not shine. It is amusing and odd that despite the gross errors in this derivation, Laplace reached the same conclusion as that mentioned above concerning black holes. We may take the argument a little further. If the density of a (spherical) star is ρ , the above condition – that the star will not release light – becomes

$$R^2 > \frac{3c^2}{8\pi G\rho};$$

so if a star (of given density) is big enough, it will not shine. As Laplace remarked, ‘il est donc possible que les plus grands corps de l’univers, soient par cela même, invisibles.’ (it is therefore possible that the largest objects in the universe are for that very reason invisible.)

5.3.2 Isotropic form of the Schwarzschild solution

For some purposes it is useful to express the Schwarzschild metric in a form that makes explicit its isotropic character. We introduce a new radial coordinate

$$\rho = \frac{1}{2}(r - m + \sqrt{r^2 - 2mr}),$$

or equivalently

$$r = \rho \left(1 + \frac{m}{2\rho} \right)^2.$$

It is then easy to see that

$$1 - \frac{2m}{r} = \frac{\left(1 - \frac{m}{2\rho} \right)^2}{\left(1 + \frac{m}{2\rho} \right)^2}$$

and that

$$\left(1 - \frac{2m}{r} \right)^{-1} dr^2 = \left(1 + \frac{m}{2\rho} \right)^4 d\rho^2,$$

so the Schwarzschild line element becomes

$$\begin{aligned} ds^2 &= - \frac{\left(1 - \frac{m}{2\rho} \right)^2}{\left(1 + \frac{m}{2\rho} \right)^2} c^2 dt^2 + \left(1 + \frac{m}{2\rho} \right)^4 (d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2 \theta d\phi^2) \\ &= - \frac{\left(1 - \frac{m}{2\rho} \right)^2}{\left(1 + \frac{m}{2\rho} \right)^2} c^2 dt^2 + \left(1 + \frac{m}{2\rho} \right)^4 (dx^2 + dy^2 + dz^2), \end{aligned} \quad (5.40)$$

where the variables x , y and z are defined by $x = \rho \sin \theta \cos \phi$, $y = \rho \sin \theta \sin \phi$, $z = \rho \cos \theta$. This is the so-called isotropic form of the metric in Schwarzschild space-time.

5.4 Time dependence and spherical symmetry: Birkhoff's theorem

Now consider a slightly more general problem; that of a spherically symmetric but time-dependent gravitational field, satisfying the vacuum field equations. For example we might have a radially pulsating star (the pulsations compatible with spherical symmetry). Then the most general form of the metric is

$$ds^2 = -P(r, t) c^2 dt^2 + Q(r, t) dr^2 + 2R(r, t) dr dt + S(r, t) r^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

As before, we can put $\hat{r} = \sqrt{S(r, t)}r$, so the final term above is $\hat{r}^2(d\theta^2 + \sin^2 \theta d\phi^2)$, and P, Q and R become replaced by new functions. Then with simple relabelling we get ($d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$)

$$ds^2 = -P(r, t) c^2 dt^2 + Q(r, t) dr^2 + 2R(r, t) dr dt + r^2 d\Omega^2. \quad (5.41)$$

Now we can find a function $f(r, t)$ such that

$$f(r, t) [P(r, t) c dt - R(r, t) dr]$$

is a *perfect differential*, $c dF(r, t)$

$$f(r, t) [P(r, t) c dt - R(r, t) dr] = c dF(r, t).$$

The condition for this is

$$\frac{\partial}{\partial r} [f(r, t) P(r, t)] = -\frac{\partial}{\partial t} [f(r, t) R(r, t)].$$

This is regarded as an equation for $f(r, t)$, which may be put in the form

$$\frac{\partial f}{\partial t} = \frac{1}{R} \left\{ \frac{\partial f}{\partial r} P + f \frac{\partial P}{\partial r} + f \frac{\partial R}{\partial t} \right\}.$$

Then, given $f(r, t_0)$ for all r at $t = t_0$, we can find $f(r, t_0 + dt)$. It then follows that

$$-\frac{1}{f^2} \frac{1}{P} c^2 dF^2 = -Pc^2 dt^2 + 2Rc dt dr - \frac{R^2}{P} dr^2$$

and hence that

$$ds^2 = -\frac{1}{f^2} \frac{1}{P} c^2 dF^2 + \left(Q + \frac{R^2}{P} \right) dr^2 + r^2 d\Omega^2. \quad (5.42)$$

Comparing this with (5.41) we note that the term in $dr dt$ has disappeared. We may now define $F(r, t)$ to be a new time parameter t'

$$t' = F(r, t);$$

then dropping the prime we may write (5.42) as

$$ds^2 = -U(r, t) c^2 dt^2 + V(r, t) dr^2 + r^2 d\Omega^2.$$

This is similar to (5.30) above, except that U and V are now functions of t as well as of r . Analogously to (5.31), then, we may put

$$ds^2 = -e^{2v(r,t)} c^2 dt^2 + e^{2\lambda(r,t)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (5.43)$$

We remind ourselves that this is the general form of the metric in a situation with spherical symmetry but time dependence also. As before, we now calculate the

connection coefficients $\Gamma^\mu_{\nu\kappa}$ and impose the vacuum field equations to find $v(r, t)$ and $\lambda(r, t)$.

In addition to the Christoffel symbols found above, it turns out that the following are non-zero:

$$\Gamma^0_{00} = \frac{1}{c} \frac{\partial v}{\partial t} = \frac{\dot{v}}{c}, \quad \Gamma^1_{01} = \Gamma^1_{10} = \frac{\dot{\lambda}}{c}, \quad \Gamma^1_{11} = \frac{\dot{\lambda}}{c} e^{2(\lambda-v)}.$$

As for the Ricci tensor, R_{22} and R_{33} are unchanged. R_{00} and R_{11} acquire the following extra terms:

$$R_{00} = \dots + \frac{1}{c^2} (\dot{\lambda}\dot{v} - \ddot{\lambda} - \dot{\lambda}^2), \quad R_{11} = \dots + \frac{1}{c^2} (\ddot{\lambda} + \dot{\lambda}^2 - \dot{\lambda}\dot{v}) e^{2(\lambda-v)}.$$

In addition, there is one non-diagonal component of $R_{\mu\nu}$:

$$R_{01} = \frac{2}{rc} \dot{\lambda}.$$

Hence the condition $R_{01} = 0$ implies that $\dot{\lambda} = 0$, so λ depends only on r ; and all the extra terms in R_{00} and R_{11} therefore vanish. The solution to the field equations is therefore found from (A), (B) and (C) above (after (5.34)), except that $v(r, t)$ is now a function of r and t . The equation $\lambda' + v' = 0$ then gives, on integration,

$$\lambda(r) + v(r, t) = \eta(t),$$

a function of t only, and (C) gives

$$(1 - 2r\lambda') e^{-2\lambda(r)} = 1 \Rightarrow e^{-2\lambda} = 1 - \frac{2m}{r}.$$

Since $\lambda(r)$ is independent of t , so is m ; $m = \text{const}$, as in the static case. Then the solution is

$$ds^2 = -e^{2\eta(t)} \left(1 - \frac{2m}{r}\right) c^2 dr^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

with $m = \frac{MG}{c^2}$. This differs from the Schwarzschild solution by the factor $e^{2\eta(t)}$ in the first term. We can, however, redefine the time coordinate: put

$$t' = \int e^{\eta(t)} dt,$$

then the first term becomes $-(1 - \frac{2m}{r})c^2 dt'^2$, and we recover the Schwarzschild metric again. We have therefore shown that *any spherically symmetric solution of the field equations is necessarily static*. This is Birkhoff's theorem. It follows that a star pulsating radially has the same external field as a star at rest – an interesting result! In other words, a radially pulsating star emits no gravitational radiation. We shall indeed see below that gravitational radiation is quadrupolar in nature; a star must oscillate in a quadrupolar manner to emit radiation – and this mode of oscillation does not possess spherical symmetry.

5.5 Gravitational red-shift

The Schwarzschild metric tensor

$$g_{\mu\nu} = \text{diag} \left[- \left(1 - \frac{2GM}{rc^2} \right), \left(1 - \frac{2GM}{rc^2} \right)^{-1}, r^2, r^2 \sin^2 \theta \right] \quad (5.44)$$

is constant in time, independent of $x^0 = ct$. The parameter t is therefore called *world time*. (Its choice is not unique; it may for instance be multiplied by an arbitrary constant.)

Consider the lapse of *proper time* τ between two events at a *fixed spatial point* in Schwarzschild space-time:

$$ds^2 = -c^2 d\tau^2 = -g_{00} c^2 dt^2, \quad g_{00}(r) = \left(1 - \frac{2GM}{rc^2} \right),$$

hence

$$d\tau = \sqrt{g_{00}} dt. \quad (5.45)$$

The element of proper time $d\tau$ is measured by a clock at the particular point, while the element of world time dt is fixed for the whole manifold. From (5.44) we see that $g_{00} < 1$, so $d\tau < dt$; in an easy slogan, clocks go slower in a gravitational field. In fact *time itself* goes slower in a gravitational field. Consider two equal, originally synchronous, clocks at some point in a field, both in an inertial frame. Now move one of them to another point, in a gravitational field, for a certain time and then bring it back to rejoin the first clock. The lapses of proper time of the clocks will not be the same; the one which has spent time in the gravitational field will be behind ('younger'). By the Equivalence Principle this is actually analogous to the twin paradox; the twin who has undergone an acceleration is younger; and an acceleration is equivalent to a gravitational field.

How are we to observe this distortion of time by a gravitational field? Suppose a particular physical process takes a certain *time* to occur, and this time is measured by a clock. The process might be a nuclear decay half-life, or it might be light emitted in a particular atomic transition, which has a particular frequency (an inverse time). We might consider measuring this characteristic time or frequency at different points in a gravitational field (where g_{00} differs) and looking for a variation. This however will not work; there will be no effect, because while the gravitational field affects the physical process it *also* affects the measuring apparatus (the clock, or the frequency measuring device), so the recorded result will always be the same.

To get an observable effect we must compare time dilation effects at two different points in the gravitational field. Let us observe, at r_1 , light coming from an atomic transition at r_2 . The wavelength of the light corresponds to a definite ds , or a definite proper time $ds \sim c d\tau$. Figure 5.2 shows the emission of two successive wave crests travelling from r_2 to r_1 . Let the world-time interval between the emission of these crests at r_2 be Δt_2 ; this is then *also* the world-time interval between the *reception* of the crests at r_1 , since the two crests of light take the same time to travel.

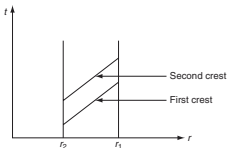


Fig. 5.2

Emission of successive crests of light, from r_2 to r_1 .

At $r = r_2$, dr_2 = proper time interval between successive crests = period of light = (frequency) $^{-1}$, so we may write

$$dr_2 = \frac{1}{\nu_2} = \frac{\lambda_2}{c} = \Delta t_2 \sqrt{g_{00}(r_2)}.$$

Also we have

$$dr_1 = \frac{1}{\nu_1} = \frac{\lambda_1}{c} = \Delta t_2 \sqrt{g_{00}(r_1)},$$

where the reader will note the same Δt_2 in both expressions, as explained above. Dividing, we have

$$\frac{\nu_1}{\nu_2} = \sqrt{\frac{g_{00}(r_2)}{g_{00}(r_1)}} \text{ with } g_{00}(r) = 1 - \frac{2GM}{rc^2}. \quad (5.46)$$

The meaning of the symbols is:

ν_1 = frequency of light emitted at r_2 and received at r_1 ,

ν_2 = frequency of light emitted and received at r_2

= frequency of light emitted and received at r_1 ,

as per the comments above. In the case of light from the Sun received on Earth we put $r = r_2$ = surface of Sun, $r = r_1$ = surface of Earth, so

ν_1 = frequency of light emitted on Sun, received on Earth,

ν_2 = frequency of light emitted and received on Sun

= frequency of light emitted and received on Earth, in the laboratory.

Since in the two cases $r = r_1$ and $r = r_2$ we have $\frac{2GM}{rc^2} \ll 1$, the expression in (5.46) may be evaluated by a binomial expansion to first order, giving

$$\begin{aligned}\frac{v_1}{v_2} &= \left(1 - \frac{2GM}{r_2 c^2}\right)^{1/2} \left(1 - \frac{2GM}{r_1 c^2}\right)^{-1/2} \\ &\approx \left(1 - \frac{GM}{r_2 c^2}\right) \left(1 + \frac{GM}{r_1 c^2}\right) \approx 1 + \frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2}\right).\end{aligned}\quad (5.47)$$

Since $r_1 > r_2$, then $v_1 < v_2$ – the light is *red-shifted*. For the Earth–Sun system,

$$r_2 = R_S = \text{solar radius} = 6.96 \times 10^8 \text{ m}$$

$$r_1 = R = \text{Earth–Sun distance} = 1.5 \times 10^{11} \text{ m} \gg R_S,$$

so to leading order we have

$$\frac{v_1}{v_2} = 1 - \frac{GM_S}{R_S c^2} \approx 1 - 2.12 \times 10^{-6}, \quad (5.48)$$

or if $v_1 = v_2 + \Delta v$, (Δv = frequency shift)

$$\frac{\Delta v}{v} = -2.12 \times 10^{-6}. \quad (5.49)$$

This is the predicted magnitude of the frequency shift of light from the Sun. Spectral lines from the Sun are compared with the ‘same’ lines observed in the laboratory. The effect is actually rather small, and difficult to detect, because broadening of spectral lines at $T = 3000$ K and Doppler shifts due to convection currents in the solar atmosphere both tend to mask the effect being looked for. Nevertheless Brault (1963) gives a value 1.05 ± 0.05 times the predicted one.

The effect is larger in white dwarfs, which have a similar mass to the Sun’s, but a radius smaller by a factor between 10 and 100, so $\frac{\Delta v}{v}$ is greater by a similar factor. The problems caused by Doppler broadening and so on are less severe, but there is a problem in estimating the mass of white dwarfs. Nevertheless, for 40 Eridani B the predicted and observed red-shifts are²

$$\text{predicted } \frac{\Delta v}{v} = -(5.7 \pm 1) \times 10^{-5}; \quad \text{observed } \frac{\Delta v}{v} = -(7 \pm 1) \times 10^{-5}.$$

A more precise test for the gravitational frequency shift, however, is not astronomical but terrestrial, and is actually a blue-shift rather than a red-shift. It is very small indeed in magnitude, but can be measured very precisely. It was performed by Pound and Rebka in 1960.³ A gamma ray from a 14.4 keV atomic transition in ^{57}Fe falls vertically in the Earth’s gravitational field through a distance of 22.6 metres. From the formulae above, with

$$g_{00}(r) = -\left(1 + \frac{2\phi(r)}{c^2}\right) \quad (5.50)$$

we have, since $\phi/c^2 \ll 1$,

² Popper (1954).

³ Pound & Rebka (1960).

$$\frac{v_1}{v_2} = \sqrt{\frac{g_{00}(r_2)}{g_{00}(r_1)}} = 1 + \frac{1}{c^2}(\phi(r_2) - \phi(r_1)). \quad (5.51)$$

Light travels from r_2 to r_1 : v_2 is the frequency measured in a laboratory at r_1 , and v_1 is the frequency of the radiation emitted at r_2 and received at r_1 . Then, with $v_1 = v_2 + \Delta v$, we have

$$\frac{\Delta v}{v} = \frac{1}{c^2}(\phi(r_2) - \phi(r_1)), \quad (5.52)$$

with $\Delta v < 0$ giving a blue-shift and $\Delta v > 0$ a red-shift. With M_E the mass of the earth and R_E its radius we have $\phi(r) = -\frac{GM_E}{r}$, $r_1 = R_E$, $r_2 = R_E + z$, where z is the height of the tower down which the gamma ray falls. Hence

$$\frac{\Delta v}{v} = \frac{GM_E}{c^2} \left(\frac{1}{R_E} - \frac{1}{R_E + z} \right) \approx \frac{GM_E z}{c^2 R_E^2} = \frac{gz}{c^2}$$

where g is the acceleration due to gravity at the Earth's surface $= 9.78 \text{ m s}^{-2}$. With $z = 22.6 \text{ m}$ this gives

$$\frac{\Delta v}{v} = 2.46 \times 10^{-15}, \quad (5.53)$$

a very small *blue-shift*. The 14.4 keV line in ^{57}Fe has a width Γ with $\Gamma/v = 1.13 \times 10^{-12}$. The width is actually small, due to the Mossbauer effect, yet nevertheless Γ/v is 460 times the value of $\Delta v/v$ above and, on the face of it, it would seem impossible to detect this blue-shift. Pound and Rebka, however, had the clever idea of superimposing onto this gravitational frequency shift a *Doppler* frequency shift, by moving the gamma ray source up and down with a velocity $v_0 \cos \omega t$. In the experiment the gamma ray is detected (at the bottom of the tower) by resonant absorption in ^{57}Fe , and the effect of moving the detector is to change the counting rate in such a way that the *gravitational* frequency shift (which is much smaller) can actually be observed.⁴ The experimental result was

$$\left(\frac{\Delta v}{v} \right)_{\text{exp}} = (2.57 \pm 0.26) \times 10^{-15}, \quad (5.54)$$

in striking confirmation of the prediction (5.53). This result has since been improved.⁵

The most precise test of the gravitational frequency shift to date, however, involves a rather different type of experiment, using a hydrogen maser in a rocket at an altitude of 10 000 km and comparing its frequency to that of a similar clock on the ground. The experiment was performed in 1976 by Vessot and Levine, who found agreement between the observed and predicted frequency shifts at the 70×10^{-6} level.⁶

Let me conclude this section with a couple of remarks. First, tests of the gravitational red-shift (or blue-shift) are in essence tests of the Equivalence Principle. They do not involve

⁴ For details the reader is referred to a good account in Weinberg (1972), pp. 82–83.

⁵ Pound & Snider (1964).

⁶ Vessot & Levine (1979), Vessot *et al.* (1980). For more details see Will (1993, 2001).

Einstein's field equations, so are not tests of General Relativity in its full form. Second, the above equations may actually be derived by treating light as a stream of photons with energy $E = h\nu$, which decreases as the light 'climbs out of' the gravitational field. This type of derivation, presented in some books, is not exactly wrong, but it should be clear that if one takes it seriously, then what is being tested is *also* quantum theory. The approach above has nothing to do with quantum theory. Light is simply described by a frequency, an inverse time. Elsewhere in General Relativity (for example the bending of its path in a gravitational field) light is described by a null geodesic.⁷

5.6 Geodesics in Schwarzschild space-time

We are interested in the motion of planets and of light in the gravitational field of the Sun, i.e. of test bodies in Schwarzschild space-time. This motion is given by the geodesic equation

$$\frac{d^2 x^\mu}{ds^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{ds} \frac{dx^\rho}{ds} = 0. \quad (5.55)$$

The connection coefficients $\Gamma^\mu_{\nu\rho}$ have already been found in (5.33) and (5.34) above. With $\lambda = -\nu$ and $e^{2\nu} = 1 - \frac{2m}{r}$ we then have

$$\begin{aligned} \Gamma^0_{10} &= \Gamma^0_{01} = \frac{m}{r(r-2m)}, \\ \Gamma^1_{11} &= -\frac{m}{r(r-2m)}, \quad \Gamma^1_{22} = -(r-2m), \quad \Gamma^1_{33} = -(r-2m) \sin^2 \theta, \\ \Gamma^2_{12} &= \Gamma^2_{21} = \Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{r}, \quad \Gamma^2_{33} = -\sin \theta \cos \theta, \\ \Gamma^3_{23} &= \Gamma^3_{32} = \cot \theta, \quad \text{others} = 0. \end{aligned} \quad (5.56)$$

With $x^0 = ct$, $i = \frac{dr}{dt}$ etc., the geodesic equation with $\mu = 0$ is

$$\ddot{t} + \frac{2m}{r(r-2m)} \dot{r} = 0$$

or

$$\frac{d}{ds} \left[\left(1 - \frac{2m}{r} \right) \dot{t} \right] = 0,$$

which is integrated to give

$$\left(1 - \frac{2m}{r} \right) \dot{t} = b = \text{const.} \quad (5.57)$$

The $\mu = 2$ geodesic equation is $\ddot{\theta} + \Gamma^2_{\kappa\lambda} \dot{x}^\kappa \dot{x}^\lambda = 0$, which with (5.56) gives

⁷ For more elaboration of this point see Okun *et al.* (2000).

$$\ddot{\theta} + \frac{2}{r} \dot{r} \dot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 = 0. \quad (5.58)$$

The $\mu = 3$ equation likewise becomes

$$\ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} = 0. \quad (5.59)$$

Instead of the $\mu = 1$ geodesic equation take the line element

$$ds^2 = -\left(1 - \frac{2m}{r}\right) c^2 dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \quad (5.60)$$

and divide it by $ds^2 = -c^2 dt^2$ to give

$$1 = \left(1 - \frac{2m}{r}\right) \dot{t}^2 - \left(1 - \frac{2m}{r}\right)^{-1} \frac{\dot{r}^2}{c^2} - \frac{r^2}{c^2} (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2). \quad (5.61)$$

The advantage of writing this equation is that for light $ds^2 = 0$ so the left hand side of (5.61) becomes 0 instead of 1. It is then possible to treat the paths of light and planets in a similar way.

Now consider a geodesic passing through a point P on the 'equator' $\theta = \pi/2$, tangent to the equatorial plane $\dot{\theta} = 0$. Then (5.58) gives $\ddot{\theta} = 0$, so $\dot{\theta}$ is *always* zero, hence θ is always $\pi/2$, and *planar motion is allowed* – in General Relativity, as in Newtonian physics. Then (5.59) gives

$$\ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} = 0$$

or

$$\frac{d}{ds} (r^2 \dot{\phi}) = 0$$

or, on integration,

$$r^2 \dot{\phi} = a = \text{const.} \quad (5.62)$$

Now substitute (5.57) and (5.62) into (5.61), noting that

$$\dot{r} = \frac{dr}{d\phi} \dot{\phi} = \frac{dr}{d\phi} \cdot \frac{a}{r^2};$$

then

$$1 = \left(1 - \frac{2m}{r}\right)^{-1} b^2 - \frac{1}{c^2} \left(1 - \frac{2m}{r}\right)^{-1} a^2 \frac{1}{r^4} \left(\frac{d\phi}{dr}\right)^2 - \frac{a^2}{c^2 r^2}.$$

Then noting that $\frac{1}{r^4} \left(\frac{d\phi}{dr}\right)^2 = \left[\frac{d}{d\phi} \left(\frac{1}{r}\right)\right]^2$, and multiplying by $\left(1 - \frac{2m}{r}\right)/a^2$ gives

$$\left[\frac{d}{d\phi} \left(\frac{1}{r}\right)\right]^2 + \frac{1}{r^2} = \frac{c^2(b^2 - 1)}{a^2} + \frac{2mc^2}{ra^2} + \frac{2m}{r^3}. \quad (5.63)$$

Differentiating with respect to ϕ gives

$$2 \frac{d}{d\phi} \left(\frac{1}{r} \right) \frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{2}{r} \frac{d}{d\phi} \left(\frac{1}{r} \right) = \frac{2mc^2}{a^2} \frac{d}{d\phi} \left(\frac{1}{r} \right) + \frac{6m}{r^2} \frac{d}{d\phi} \left(\frac{1}{r} \right).$$

Rejecting the solution $\frac{d}{d\phi} \left(\frac{1}{r} \right) = 0$, which is a circle, we find

$$\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = \frac{mc^2}{a^2} + \frac{3m}{r^2}, \quad (5.64)$$

which is the differential equation for the orbit. In the case of light the left hand side of (5.61) is 0 instead of 1, and (5.64) becomes replaced by

$$\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = \frac{3m}{r^2}. \quad (5.65)$$

In these equations it will be recalled that $m = \frac{GM}{c^2}$, M being the mass of the gravitating body.

5.7 Precession of planetary orbits

In Newtonian theory the differential equation for a planet moving in a non-circular orbit is

$$\frac{d^2 u}{d\phi^2} + u = \frac{1}{p} \quad (5.66)$$

where $u = 1/r$ and $p = a^2/GM = a^2/mc^2$ (see Problem 5.3). The quantity p , the semi-latus rectum of the ellipse, is given by

$$p = a_0(1 - e^2)$$

where a_0 is the semi major axis and e the eccentricity. The general solution to Equation (5.66) is

$$u = \frac{1}{r} = \frac{1}{p} [1 + e \cos(\phi - \alpha)],$$

where e and α are constants of integration. We may choose e to be non-negative ($e \geq 0$), since the cosine term changes sign when $\alpha \rightarrow \alpha + \pi$. In addition a suitable rotation of the coordinate system allows us to choose $\alpha = 0$, so the solution above becomes

$$u = \frac{1}{r} = \frac{1}{p} (1 + e \cos \phi), \quad e \geq 0. \quad (5.67)$$

This is the equation for an ellipse with eccentricity e . In the case $e = 0$ the ellipse becomes a circle. The particle's closest approach is at $\phi = 0$: this is called the perihelion if M is the Sun (and the perigee if M is the Earth).

The general relativistic equation for the orbit, (5.64), has a correction term added to the Newtonian equation (5.66). The relative magnitude of this term, in the case of Mercury, where $r \sim 6 \times 10^7$ km, is

$$\frac{3m}{r^2} / \frac{1}{r} = \frac{3m}{r} \sim 10^{-7}.$$

For the last term in (5.64) we then substitute (5.67) and neglect the term in e^2 , giving

$$\begin{aligned} \frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} &= \frac{mc^2}{a^2} + \frac{3m^3 c^4}{a^4} (1 + 2e \cos \phi) \\ &\approx \frac{mc^2}{a^2} + \frac{6m^3 c^4}{a^4} e \cos \phi. \end{aligned} \quad (5.68)$$

The solution to (5.68) is the solution to (5.66) (which is (5.67)) plus the solution to

$$\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = \frac{6m^3 c^4}{a^4} e \cos \phi,$$

that is,

$$\frac{1}{r} = \frac{mc^2}{a^2} (1 + e \cos \phi) + \frac{6em^3 c^4}{a^4} \phi \sin \phi$$

(where it may be noted that the last term is *non-periodic*)

$$\begin{aligned} &= \frac{mc^2}{a^2} \left(1 + e \cos \phi + \frac{3em^2 c^2}{a^2} \phi \sin \phi \right) \\ &= \frac{mc^2}{a^2} \left[1 + e \cos \left\{ \phi \left(1 - \frac{3m^2 c^2}{a^2} \right) \right\} \right] + O\left(\frac{m}{r}\right)^2. \end{aligned}$$

The perihelion occurs when $1/r$ is a maximum, that is when $\cos \left[\phi \left(1 - \frac{3m^2 c^2}{a^2} \right) \right] = 1$, hence

$$\begin{aligned} \phi &= 0, \frac{2\pi}{1 - \frac{3m^2 c^2}{a^2}}, \frac{4\pi}{1 - \frac{3m^2 c^2}{a^2}}, \dots \\ &\approx 0, 2\pi + \frac{6\pi m^2 c^2}{a^2}, \dots \equiv 0, 2\pi + \delta\phi, \dots \end{aligned}$$

where $\delta\phi$ is the *precession* of the ellipse in one revolution:

$$\delta\phi = \frac{6\pi m^2 c^2}{a^2} = \frac{6\pi m}{p} = \frac{6\pi MG}{c^2 a_0 (1 - e^2)} \approx 0.1'' \quad (5.69)$$

where M is the mass of the Sun and a_0 the semi-major axis of Mercury $\approx 5.8 \times 10^{10}$ m and $e = 0.21$. This is only a tiny amount, but the precession is cumulative and in one hundred *Earth* years

$$\delta\phi_{100} = 43.03''. \quad (5.70)$$

An exaggerated picture of a precessing elliptical orbit is shown in Fig. 5.3.

The total observed precession of Mercury is $5600.73 \pm 0.41''$ per century. The Newtonian-calculated precession, caused by the motion of the other planets (of which Venus, Earth and

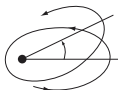


Fig. 5.3

A precessing elliptical orbit.

Jupiter give the largest contributions) is $5557.62 \pm 0.4''$.⁸ The discrepancy between these figures is 43.11 ± 0.4 . This is consistent with (5.70) and the effect is therefore well explained by General Relativity.

5.8 Deflection of light

We use Equations (5.57)–(5.59), but (5.61) has zero on the left hand side when light, rather than a massive test body, is being considered. Then the equation for the ‘orbit’ is (5.65):

$$\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = \frac{3m}{r^2}. \quad (5.65)$$

where, as before, $3m/r^2 \gg 1/r$. The solution to $\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = 0$ is

$$\frac{1}{r} = \frac{1}{r_0} \cos \phi, \quad (5.70)$$

a straight line (Fig. 5.4). Equation (5.65) is solved by substituting (5.70) into the right hand side:

$$\frac{d^2}{d\phi^2} \left(\frac{1}{r} \right) + \frac{1}{r} = \frac{3m}{r_0^2} \cos^2 \phi. \quad (5.71)$$

This has the particular solution

$$\frac{1}{r} = \frac{m}{r_0^2} (1 + \sin^2 \phi), \quad (5.72)$$

so (5.65) has the general solution

$$\frac{1}{r} = \frac{1}{r_0} \cos \phi + \frac{m}{r_0^2} (1 + \sin^2 \phi). \quad (5.73)$$

Now consider the asymptotes:

$$(5.70): r \rightarrow \infty, \quad \phi \rightarrow \pm \pi/2; \quad (5.73): r \rightarrow \infty, \quad \phi \rightarrow \pm (\pi/2 + \delta)$$

⁸ These figures are taken from Robertson & Noonan (1968), p. 239.

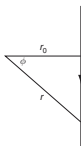


Fig. 5.4

A straight line – the path of a light ray in flat space-time.

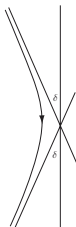


Fig. 5.5

The path of a light ray in a Schwarzschild field.

with

$$-\frac{1}{r_0} \sin \delta + \frac{m}{r_0^2} (1 + \cos^2 \delta) = 0 \Rightarrow \delta = \frac{2m}{r_0},$$

since δ is small. Hence the path of the light ray is as shown in Fig. 5.5 and the total deflection is, in the case of the Sun,

$$\Delta = 2\delta = \frac{4m}{r_0} = \frac{4MG}{r_0 c^2}. \quad (5.74)$$

For light just grazing the Sun the distance of nearest approach, r_0 , is effectively the Sun's radius R , and

$$\Delta = \frac{4MG}{Rc^2} = 1.75''. \quad (5.75)$$

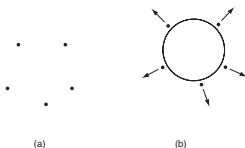


Fig. 5.6

(a) Stars in the night sky; (b) the same stars seen near the Sun's edge at a time of total solar eclipse – the Sun's light is blocked by the Moon.

Stars seen close to the Sun's edge are of course not visible by day, but become visible at the time of a total eclipse. Their position relative to the background of other stars then appears shifted relative to what it is in the usual night sky – they appear to move out from the Sun, as in Fig. 5.6.

This prediction of General Relativity was first verified in 1919. Two separate expeditions, to Brazil and Guinea, reported deflections of $1.98 \pm 0.16''$ and $1.61 \pm 0.40''$, in reasonable accord with Einstein's prediction. It was this 'bending of light' that really made Einstein famous. Many measurements of Δ were made in succeeding years, but the accuracy did not really increase until the advent of very long baseline (more than 800 km) radio interferometry in 1972, using quasar sources. A 1995 measurement on 3C273 and 3C279 gives⁹

$$\frac{\Delta_{\text{observed}}}{\Delta_{\text{predicted}}} = 0.9996 \pm 0.0017. \quad (5.76)$$

5.9 Note on PPN formalism

Since Einstein proposed the theory of General Relativity various other people have proposed modifications of it. They have in common with it that they are all *geometric* theories of gravity, i.e. the gravitational field is manifested in the geometry of space-time itself – in particular it is curved. These theories then give, in the case of a static, spherically symmetric mass, a metric similar to the Schwarzschild metric, but with some different numerical coefficients. A convenient way of discussing the results of these theories is to introduce two parameters β , γ into the Schwarzschild solution, so that it now has the form

⁹ Lebach *et al.* (1995): see also Will (2001) for more information. The result quoted is cast in the language of the PPN formalism, which is explained in the next section.

$$ds^2 = -\left(1 - \frac{2m}{r} + 2(\beta - \gamma) \frac{m^2}{r^2}\right) c^2 dt^2 + \frac{dr^2}{1 - \frac{2\gamma m}{r}} + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (5.77)$$

In General Relativity

$$\text{GR: } \gamma = 1, \quad \beta = 1. \quad (5.78)$$

Einstein's theory and its rivals are all post-Newtonian theories, and these parameters γ, β are then post-Newtonian parameters.

The most commonly cited alternative to General Relativity is the Brans–Dicke theory, which includes a scalar field as a way of incorporating Mach's Principle. This theory has, however, together with all the other rivals to General Relativity, been to all intents and purposes ruled out on empirical grounds.¹⁰ Many writers, nevertheless, pursue an interest in these theories, and the PPN (parametrised post-Newtonian) formalism does actually provide a convenient way of presenting the results of tests of General Relativity. For example the perihelion precession of a planetary orbit is

$$\delta\phi = \left(\frac{2 + 2\gamma - \beta}{3}\right) \frac{6\pi m^2 c^2}{a^2} \quad (5.79)$$

which reduces to (5.69) when $\beta = \gamma = 1$. In a similar way the deflection of light is given by

$$\Delta = \left(\frac{1 + \gamma}{2}\right) \frac{4m}{r_0} \quad (5.80)$$

which reduces to (5.74) when $\gamma = 1$.

5.10 Gravitational lenses

The power of large masses to bend light paths invites comparison with optical lenses – hence the topic of ‘gravitational lenses’. The analogy, however, is not complete, as we shall see.

First, consider light coming from infinity and bent by a ‘lens’ L to reach an observer O . Call the distance of closest approach between the light path and the lens b – an ‘impact parameter’. The bending of the light path may be simply represented as taking place at one point, discontinuously, as shown in Fig. 5.7. (Note that the angles involved here are extremely small; of the order of a few seconds of arc at most, as we saw above. The diagram is therefore greatly exaggerated.) Light is detected by the observer at a distance D from L :

$$D = \frac{b}{\theta}. \quad (5.81)$$

However from (5.74) above $\theta = \Delta = \frac{4m}{b}$ so

¹⁰ Some years ago there was a problem connected with solar oblateness, and reason to hope that the Brans–Dicke theory might clear this up. It now appears, though, that the problem has probably gone away. See for example d’Inverno (1992), p. 206.

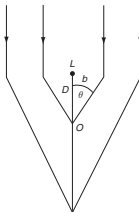


Fig. 5.7

Light coming from infinity is bent by a lens L to reach an observer O .

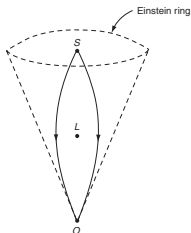


Fig. 5.8

In a symmetrical configuration light from a source S is bent by a lens L to reach an observer O , who then sees the source as an 'Einstein ring'.

$$D = \frac{b^2}{4m}, \quad (5.82)$$

where, as always, $m = MG/c^2$, M being the mass of the lens. D therefore increases with b : rays 'further out' from L reach the axis at a different point. The lens *does not* focus the rays to one point.

Consider the symmetric situation where the source S , lens L and observer O are in a straight line, as in Fig. 5.8. In a planar cross section the source S appears to the observer O as two images. There is cylindrical symmetry, however, so the image is actually a ring – the

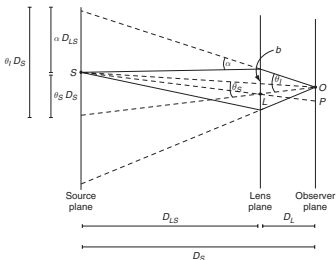


Fig. 5.9

Observer O sees light from the source S bent by the lens L : O , S and L are not in a straight line.

‘Einstein ring’. What is its (angular) radius? The general situation is sketched in Fig. 5.9: here the source S , lens L and observer O are not in a straight line. The distances between the planes are as marked: D_{LS} is the distance from the lens to the source, etc. These distances depend on the angles, also as marked. Then

$$\theta_I D_S = \theta_S D_S + \alpha D_{LS} \quad (5.83)$$

but (see (5.74))

$$\alpha = \frac{4m}{b} = \frac{4m}{\theta_I D_L}$$

so

$$\theta_I = \theta_S + \frac{4}{\theta_I} \frac{m D_{LS}}{D_L D_S},$$

which on relabelling $\theta_I \rightarrow \theta$ becomes

$$\theta = \theta_S + \frac{\theta_E^2}{\theta}, \quad (5.84)$$

where

$$\theta_E^2 = \frac{4m D_{LS}}{D_L D_S}, \quad (5.85)$$

and θ_E is called the ‘Einstein angle’. The solutions to this equation give the angular positions of images in the sky – there are generally two solutions. In the situation where $D_{LS}, D_S \gg D_L$ (the source is at an ‘infinite’ distance) we have

$$\theta_E^2 \approx \frac{4m}{D_L}. \quad (5.86)$$

The significance of the Einstein angle is that in the symmetric case, when the source, lens and observer are collinear, $\theta_S = 0$ so $\theta = \theta_E$: the image is a ring with this angular radius. In the case of lensing in the galaxy (of one star by another, not the Sun), $\theta_E \approx 10^{-3}''$, an angle which is too small to be resolvable by current telescopes. This is an example of *microlensing*. On the cosmological scale, however, (the lensing of one galaxy by another) $\theta_E \approx 1''$, which is resolvable, and indeed there is observational evidence for lensing on this scale, known as *macrolensing*. (For the figures quoted above, see Problem 5.4.)

In a cosmological context lensing may be used – at least in principle, and practice seems now not far behind – to find the masses and distances away of galaxies acting as lenses. This is important information, for example in the study of dark matter; the determination of masses by this method is *non-dynamical*, in contrast with the usual method of detecting dark matter. It will be shown here, in a very simple (and somewhat unrealistic) model how this works.

In the approximation $D_{LS} \sim D_S \gg D_L = D$ Equation (5.84) holds with

$$\theta_E^2 = \frac{4m}{D} \quad (5.87)$$

where $m = MG/c^2$, M being the mass of the lens, and D being its distance away. Hence

$$\theta^2 - \theta_S \theta - \theta_E^2 = 0. \quad (5.88)$$

This equation has two solutions, θ_1 and θ_2 (which are of different signs). Their product is

$$|\theta_1 \theta_2| = \theta_E^2 = \frac{4m}{D}. \quad (5.89)$$

We now find one more relation between θ_1 and θ_2 . This depends on the differing *path lengths* of the two rays. Consider a point P in the observer plane on the line SL extrapolated, as in Fig. 5.10. From P an Einstein ring is seen: so in the planar diagram light fronts arriving

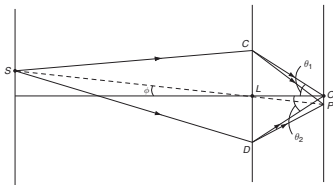


Fig. 5.10

Light rays from the source S to observer O .

along the two paths do so at the same time. In addition, to a good approximation, α , the angle between these two rays, is equal to $\theta_1 + \theta_2$, the corresponding angle at O . Let $OP = d$, $CP = l_1$, $DP = l_2$. Also let $CL = LD = h$ (and $LO = D$, above). Then by Pythagoras

$$l_1 = \sqrt{D^2 + (h + d)^2} = D \left[1 + \frac{(h + d)^2}{D^2} \right]^{1/2} \approx D + \frac{(h + d)^2}{2D}.$$

Similarly

$$l_2 \approx D + \frac{(h - d)^2}{2D},$$

so

$$l_1 - l_2 = \frac{2hd}{D} = 2h\phi.$$

But $2h = D(\theta_1 + \theta_2)$, so $l_1 - l_2 = D(\theta_1 + \theta_2)\phi$ and the time difference between the two signals arriving at P is

$$\Delta t = \frac{l_1 - l_2}{c} = \frac{D}{c}(\theta_1 + \theta_2)\phi. \quad (5.90)$$

On the other hand $\phi \approx \theta_S$ (compare Figs. 5.9 and 5.10) and from the theory of quadratic equations we have from (5.88)

$$|\theta_1 - \theta_2| = \theta_S$$

(recall once more that θ_1 and θ_2 are of different signs), hence $\phi = |\theta_1 - \theta_2|$ and (5.90) gives

$$\Delta t = \frac{D}{c} |\theta_1^2 - \theta_2^2|, \text{ hence}$$

$$|\theta_1^2 - \theta_2^2| = \frac{c \Delta t}{D}. \quad (5.91)$$

This is a second relation between θ_1 and θ_2 ; so by measuring θ_1 , θ_2 and Δt , the Equations (5.89) and (5.91) allow us in principle to determine D and m , the distance away and mass of the lens. It was mentioned above that this model is too simplistic; two effects which have been ignored are the echo delay effect (the subject of the next section) and the expansion of the Universe. Nevertheless we have shown how gravitational lensing is a useful tool for cosmology.

5.11 Radar echoes from planets

A previous test for General Relativity, that of the bending of light paths, is a test of the *orbit* of a light ray. Modern techniques, however, allow the possibility of measuring directly the *time* for light to travel from a source to a reflector (planet or space probe) and back. In appropriate circumstances this differs from the Newtonian expression by an amount of the order of 10^{-4} seconds, an easily measurable time. The best reflector is a planet (which is less subject to non-gravitational forces like the solar wind than a space

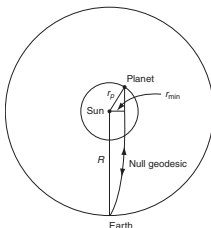


Fig. 5.11

Radar travelling from the Earth to a planet and back, with $r_{\min}$ the distance of closest approach to the Sun.

probe is), and the situation in which relativistic effects are greatest is when the planet is in 'superior conjunction', i.e. the Sun is almost on a straight line joining the Earth and the planet. This is the situation sketched in Fig. 5.11. General relativity predicts that the echo is *delayed* relative to the Newtonian prediction, that is, relative to the signal travelling out and back in a straight line.

Let $t(r_1, r_2)$ be the coordinate time between the emission of a light signal at $r = r_1$ and its reception at $r = r_2$. Then the coordinate time interval between the emission of a signal from the Earth and reception of the echo is

$$T = 2 [t(R, r_{\min}) + t(r_p, r_{\min})] \quad (5.92)$$

where $r_{\min}$ is the distance of closest approach of the path to the Sun – see Fig. 5.11. The corresponding *proper time* interval is

$$\tau = T \sqrt{1 - \frac{2GM}{rc^2}},$$

and since $\frac{GM}{rc^2} T$ is of the order of 10^{-8} s, we ignore this correction factor; we shall see below that the time delay is of the order of 100 μ s, so the dominant effect is due to the fact that the path is not a straight line. If light travelled in straight lines we should have $t(r, r_{\min}) = t_E(r, r_{\min})$ (E stands for Euclidean), with

$$t_E(r, r_{\min}) = \frac{1}{c} \sqrt{r^2 - r_{\min}^2}, \quad (5.93)$$

by Pythagoras.

To obtain $t(r, r_{\min})$ in the relativistic case we return to the Schwarzschild metric (5.36). Put $ds^2 = 0$ and $d\theta = 0$ (motion of light in a plane); then we have

$$0 = -\left(1 - \frac{2m}{r}\right) + \frac{1}{c^2} \left[\frac{\left(\frac{dr}{dt}\right)^2}{1 - \frac{2m}{r}} + r^2 \left(\frac{d\phi}{dt}\right)^2 \right]. \quad (5.94)$$

We can eliminate $\frac{d\phi}{dt}$ by using Equations (5.57) and (5.62), from which

$$\frac{r^2 \frac{d\phi}{d\tau}}{\left(1 - \frac{2m}{r}\right) \frac{dt}{d\tau}} = \frac{r^2}{\left(1 - \frac{2m}{r}\right)} \frac{d\phi}{dt} = \frac{a}{b} = B = \text{const},$$

and hence

$$\frac{d\phi}{dt} = \frac{B}{r^2} \left(1 - \frac{2m}{r}\right).$$

Then (5.94) becomes

$$0 = -\left(1 - \frac{2m}{r}\right) + \frac{1}{c^2} \left[\frac{\left(\frac{dr}{dt}\right)^2}{1 - \frac{2m}{r}} + \frac{B^2}{r^2} \left(1 - \frac{2m}{r}\right)^2 \right]. \quad (5.95)$$

We can find B by noting that $\frac{dr}{dt} = 0$ when $r = r_{\min}$, since then the velocity has no radial component (see the diagram). This gives

$$B^2 = \frac{c^2 r_{\min}^2}{\left(1 - \frac{2m}{r_{\min}}\right)}$$

and (5.95) gives, after a bit of rearrangement,

$$\frac{dr}{dt} = c \left(1 - \frac{2m}{r}\right) \left[1 - \frac{r_{\min}^2 \left(1 - \frac{2m}{r}\right)^2}{r^2 \left(1 - \frac{2m}{r_{\min}}\right)} \right]^{1/2}.$$

Then

$$t(r, r_{\min}) = \frac{1}{c} \int_{r_{\min}}^r \frac{dr'}{\left(1 - \frac{2m}{r'}\right) \left[1 - \frac{r_{\min}^2 \left(1 - \frac{2m}{r'}\right)^2}{r'^2 \left(1 - \frac{2m}{r_{\min}}\right)} \right]^{1/2}}. \quad (5.96)$$

This is the time taken for light to travel from $r_{\min}$ to r . To first order in the small quantities $\frac{m}{r}$ and $\frac{m}{r_{\min}}$ this gives

$$t(r, r_{\min}) \approx \frac{1}{c} \int_{r_{\min}}^r \frac{r' dr'}{\sqrt{r'^2 - r_{\min}^2}} \left[1 + \frac{2m}{r'} + \frac{mr_{\min}}{r'(r' + r_{\min})} \right], \quad (5.97)$$

and on integration

$$t(r, r_{\min}) = \frac{1}{c} \left[\sqrt{r^2 - r_{\min}^2} + 2m \ln \left(\frac{r + \sqrt{r^2 - r_{\min}^2}}{r_{\min}} \right) + m \sqrt{\frac{r - r_{\min}}{r + r_{\min}}} \right]. \quad (5.98)$$

(Confirmation of the above expressions is left to Problems 5.5 and 5.6.) Note that the first term in (5.98) is the Euclidean expression (5.93); and note also that the extra terms are *positive*: t is *greater* than in the flat space result – light travels in a curve.

We can then state that the time for light to travel from the Earth (at $r = R$) to $r = r_{\min}$ is, since $R \gg r_{\min}$, $r_{\min}$ being taken to be of the order of the Sun's radius,

$$t(R, r_{\min}) = t_1 = \frac{1}{c} \left[\sqrt{R^2 - r_{\min}^2} + 2m \ln \left(\frac{2R}{r_{\min}} \right) + m \right]. \quad (5.99)$$

Similarly the time from $r = r_{\min}$ to the planet at $r = r_p$ is

$$t(r_p, r_{\min}) = t_2 = \frac{1}{c} \left[\sqrt{r_p^2 - r_{\min}^2} + 2m \ln \left(\frac{2r_p}{r_{\min}} \right) + m \right], \quad (5.100)$$

and the total time of travel from the Earth back to the Earth is

$$\begin{aligned} T &= 2(t_1 + t_2) \\ &= \frac{2}{c} \left[\sqrt{R^2 - r_{\min}^2} + \sqrt{r_p^2 - r_{\min}^2} + 2m \ln \left(\frac{4Rr_p}{r_{\min}^2} \right) + 2m \right] \\ &\approx \frac{2}{c} \left[R + r_p + 2m \left\{ \ln \left(\frac{4Rr_p}{r_{\min}^2} \right) + 1 \right\} \right]. \end{aligned} \quad (5.101)$$

It is instructive to calculate this quantity for a specific case: let us take Mars, whose distance from the Sun is $r_p = 1.52 \text{ AU} = 2.28 \times 10^{11} \text{ m}$. Then

$$\frac{2}{c}(R + r_p) = 2.52 \times 10^3 \text{ s} \approx 42 \text{ min}. \quad (5.102)$$

This is the 'Euclidean' time. The excess time, or delay, is clearly greatest when $r_{\min}$ takes its smallest value, which is the radius of the Sun R_S . In that case

$$\frac{4Rr_p}{R_S^2} = 2.82 \times 10^5, \quad \ln \left(\frac{4Rr_p}{R_S^2} \right) = 12.55$$

and the 'echo delay' is

$$\begin{aligned} \Delta T &\equiv \frac{4m}{c} \left[\ln \left(\frac{4Rr_p}{R_S^2} \right) + 1 \right] \\ &= 2.66 \times 10^{-4} \text{ s} = 266 \mu\text{s}. \end{aligned} \quad (5.103)$$

It is clear that to detect this effect an accuracy of 1 part in 10^7 is necessary: to measure it to within 1%, measurement is required to 1 part in 10^9 . Atomic clocks give an accuracy of 1

part in 10^{12} , so this is achievable, but it means on the other hand that $R + r_p$ must be known to an accuracy of the order of 1 km, which presents something of a challenge. Nevertheless the results are impressive; and are often cast in the PPN formalism. In that formalism the echo delay above is

$$\Delta T = \left(\frac{1+\gamma}{2} \right) \frac{4m}{c} \left[\ln \left(\frac{4Rr_p}{R_S^2} \right) + 1 \right], \quad (5.104)$$

where, of course (see (5.78)), $\frac{(1+\gamma)}{2} = 1$ in General Relativity. Will (2001) displays the measured values of $\frac{(1+\gamma)}{2}$ and concludes that agreement with General Relativity holds to 0.1 per cent.

5.12 Radial motion in a Schwarzschild field: black holes – frozen stars

As observed already, the form of the Schwarzschild metric (5.36) suggests that there may be interesting phenomena associated with the surface $r = 2GM/c^2 = 2m$. In this section we consider the radial fall of objects in a Schwarzschild field and see the first indication of the bizarre nature of this surface.

Consider the motion of a particle falling radially in a Schwarzschild field – into a star, for example. Suppose it starts at $r = R$ with $\frac{dr}{dt} = 0$, as shown in Fig. 5.12, where the spherical surface $r = 2m$ is also marked. For radial motion we have

$$ds^2 = - \left(1 - \frac{2m}{r} \right) c^2 dt^2 + \left(1 - \frac{2m}{r} \right)^{-1} dr^2.$$

With $ds^2 = -c^2 d\tau^2$, $\dot{t} = \frac{dt}{d\tau}$, $\dot{r} = \frac{dr}{d\tau}$ this gives

$$\left(1 - \frac{2m}{r} \right) \dot{t}^2 - \frac{1}{c^2} \left(1 - \frac{2m}{r} \right)^{-1} \dot{r}^2 = 1, \quad (5.105)$$

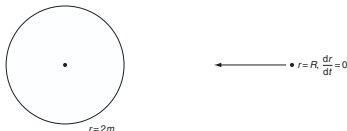


Fig. 5.12

A particle falling radially in a Schwarzschild field.

which, on putting $\dot{r} = \frac{dr}{dt} i$ gives

$$\left[c^2 \left(\frac{r-2m}{r} \right) - \left(\frac{r}{r-2m} \right) \left(\frac{dr}{dt} \right)^2 \right] i^2 = c^2. \quad (5.106)$$

The boundary condition $r=R, \frac{dr}{dt} = 0$ gives

$$\left(\frac{dt}{dr} \right)_{r=R} = \left(\frac{R}{R-2m} \right)^{1/2}. \quad (5.107)$$

From (5.57) above

$$\left(1 - \frac{2m}{r} \right) i = b = \text{const},$$

hence

$$b = \left(1 - \frac{2m}{R} \right) \left(\frac{dt}{dr} \right)_R = \left(\frac{R-2m}{R} \right)^{1/2}$$

and therefore

$$\frac{dt}{dr} = \left(\frac{r}{r-2m} \right) b = \left(\frac{r}{r-2m} \right) \left(\frac{R-2m}{R} \right)^{1/2}. \quad (5.108)$$

Substituting this into (5.106) gives, on rearrangement

$$\frac{dr}{dt} = -c \frac{(r-2m)(2m)^{1/2}(R-r)^{1/2}}{r^{3/2}(R-2m)^{1/2}} \quad (5.109)$$

Note the minus sign, selected on taking the square root: it reflects the fact that we are considering a 'fall' into the star, so r decreases as t increases. Finally we have

$$t = -\frac{1}{c} \left(\frac{R-2m}{2m} \right)^{1/2} \int_R^r \frac{\rho^{3/2} d\rho}{(\rho-2m)(R-\rho)^{1/2}}. \quad (5.110)$$

This is the time taken for a particle to travel from $r=R$ to an arbitrary r , as measured in Schwarzschild coordinates, in which t is the time parameter, which is the time measured on a clock 'at infinity' – by a distant observer. It is clear that the form of the integrand above will diverge when $\rho \rightarrow 2m$. To investigate this, put

$$\rho = 2m + \varepsilon$$

with ε small. Then (5.109) gives

$$\begin{aligned} ct &= - \left(\frac{R-2m}{2m} \right)^{1/2} \int_{R-2m}^{r-2m} \frac{(2m+\varepsilon)^{3/2} d\varepsilon}{\varepsilon(R-2m)^{1/2}} = -2m \int_{R-2m}^{r-2m} \frac{d\varepsilon}{\varepsilon} \\ &= -2m \ln \left(\frac{r-2m}{R-2m} \right), \end{aligned}$$

or

$$r - 2m = (R - 2m) \exp(-ct/2m). \quad (5.111)$$

So, as $t \rightarrow \infty$, $r \rightarrow 2m$: the radial infall of a particle to the Schwarzschild radius takes an *infinite* time as seen from a safe distance. This is always assuming, of course, that the surface $r = 2m$ is *outside* the star, in the vacuum. If we transfer this conclusion to the outer layers of a collapsing star itself, we then learn that the collapse of a star, as seen from the outside, takes an infinite time to reach a *finite* size. The star seems to collapse more and more slowly, and is *never* seen collapsing to a point. Zel'dovich and Novikov (1996) call this phenomenon a 'frozen star'.

This describes the fall of a particle (or of the star itself) as seen by a distant observer. What about the fall of the particle measured by its own clock? We need to calculate the *proper time* for the particle to reach $r = 2m$, and therefore want $\frac{dr}{d\tau}$. From (5.108) and (5.109)

$$\frac{dr}{d\tau} = \frac{dr}{dt} \cdot \frac{dt}{d\tau} = -c \left(\frac{2m(R-r)}{Rr} \right)^{1/2} = -c \left(\frac{2m}{R} \right)^{1/2} \left(\frac{R}{r} - 1 \right)^{1/2}$$

so that

$$\tau = -\frac{1}{c} \sqrt{\frac{R}{2m}} \int_R^r \frac{dr'}{\left(\frac{R}{r'} - 1\right)^{1/2}}.$$

On putting $\rho = \frac{r}{R}$ this gives

$$\tau = -\frac{1}{c} \sqrt{\frac{R^3}{2m}} \int_1^{\rho} \frac{d\rho'}{\left(\frac{1}{\rho'} - 1\right)^{1/2}}.$$

The integral is straightforward to perform and we find

$$\tau = \frac{1}{c} \sqrt{\frac{R^3}{2m}} \left[\sqrt{\rho(1-\rho)} + \cos^{-1} \sqrt{\rho} \right]. \quad (5.112)$$

This is the proper time for collapse from $\frac{r}{R} = \rho = 1$ down to an arbitrary value of ρ (< 1). It is clearly *finite* for all r , even down to $r = 0$. In fact the proper time to reach $r = 0$ is

$$\tau_0 = \frac{\pi}{2c} \sqrt{\frac{R^3}{2m}}. \quad (5.113)$$

So in summary: an object falls into a collapsed star. Viewed from the outside, the object may still be seen, even after an infinite time, as it asymptotically approaches the Schwarzschild radius $r = 2m$. Measured on its own clock, however, the object only exists for a time given by (5.113), after which it is crushed to unlimited density at the centre of the star. It seems as if nothing strange happens at $r = 2m$.¹¹ In a scenario like this General Relativity provides a spectacular illustration of the relativity of time.

¹¹ It may be, however, that the scalar quantity constructed by contracting the covariant derivative of the Riemann tensor with itself changes sign at $r = 2m$, so that in principle a detector (accelerometer) with sufficient sensitivity should be able to register when the Schwarzschild surface is crossed. I am grateful to Brian Steadman for this remark.

5.13 A gravitational clock effect

After the rather lofty topics above, consider the following simpler, but equally interesting, experiment. Suppose there are two synchronised atomic clocks on the equator. One is placed on board an aircraft which flies once round the Earth (at a height h) while the other remains on the ground. After the flight, what is the time discrepancy between the clocks?¹²

We use the Schwarzschild metric on the equator ($\theta = \pi/2$, $r = \text{const}$), so that (5.36) gives

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right)c^2 dt^2 + r^2 d\phi^2 = -c^2 d\tau^2 \quad (5.114)$$

where τ is proper time. Referring to Fig. 5.13, observer A remains at rest on the Earth, rotating with angular velocity $\omega = \frac{d\phi}{dt}$; hence $d\phi = \omega dt$ and from above

$$d\tau_A^2 = \left[\left(1 - \frac{2GM}{Rc^2}\right) - \frac{R^2\omega^2}{c^2} \right] dt^2;$$

and since $\frac{2GM}{c^2} \ll 1$, $\frac{R^2\omega^2}{c^2} \ll 1$, then

$$d\tau_A \approx \left[1 - \frac{GM}{Rc^2} - \frac{R^2\omega^2}{2c^2} \right] dt. \quad (5.115)$$

(Note the logic implied by this derivation: A is moving in a static Schwarzschild field with velocity $R\omega = R\frac{d\phi}{dt}$.) Observer B, on the other hand, flies round the Earth at a height h : as shown in Fig. 5.12 she flies eastward. She moves at a speed v relative to the ground, and so with a speed $\approx v + (R+h)\omega$ with respect to the 'static' frame of the Schwarzschild field. Then

$$d\tau_B^2 = \left[1 - \frac{2GM}{(R+h)c^2} - \frac{\{(R+h)\omega + v\}^2}{c^2} \right] dt^2$$

and

$$d\tau_B \approx \left[1 - \frac{GM}{(R+h)c^2} - \frac{R^2\omega^2 + 2R\omega v + v^2}{2c^2} \right] dt. \quad (5.116)$$

Hence

$$d\tau_A - d\tau_B = \left[-\frac{GMh}{R^2c^2} + \frac{(2R\omega + v)v}{2c^2} \right] dt. \quad (5.117)$$

¹² The following treatment follows closely that of Berry (1976), Section 5.2.

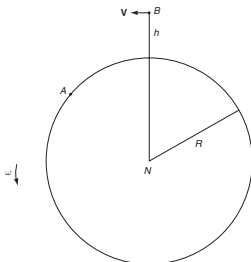


Fig. 5.13

The Earth, viewed from above: N is the North Pole. Observer A is at rest on the equator, while B travels with speed v at a height h above the Earth in the equatorial plane. They each carry a clock.

Then, defining the quantity

$$\Delta = \frac{d\tau_A - d\tau_B}{d\tau_A}, \quad (5.118)$$

we have

$$\Delta \approx -\frac{GMh}{R^2 c^2} + \frac{(2R\omega + v)v}{2c^2}. \quad (5.119)$$

Putting in numbers, with $h = 10^4$ m, $v = 300$ m s⁻¹, $\frac{GM}{R^2} = g = 9.81$ m s⁻², then $\frac{gh}{c^2} = 1.09 \times 10^{-12}$, $2R\omega = 931$ m s⁻¹ and $\frac{(2R\omega + v)v}{2c^2} = 2.1 \times 10^{-12}$ and

$$\Delta_{\text{Eastward}} = 1.0 \times 10^{-12}; \quad (5.120)$$

‘eastward’ because this is what we are considering. With a *westward* journey $v \rightarrow -v$ and $\frac{(2R\omega - v)v}{2c^2} = 1.05 \times 10^{-12}$, giving

$$\Delta_{\text{Westward}} = -2.1 \times 10^{-12}. \quad (5.121)$$

These quantities are measurable: the fractional accuracy of cesium clocks is about 1 part in 10^{13} . In 1971 Hafele and Keating¹³ placed very accurate cesium clocks aboard commercial airliners and verified the above predictions to about 10%. (The orbits were not in the

¹³ Hafele & Keating (1972).

equatorial plane, but adjustments were made for this.) This clock effect is a gravitational analogue of the twin paradox, as well as being a test for the Schwarzschild solution.

Further reading

The ‘problem of motion’ is treated in Bergmann (1942), Chapter 15. A later review of it appears in Einstein & Infeld (1949), reprinted in Kilmister (1973). See also Bażański (1962) and Synge (1964), Section 4.6.

A translation of Laplace’s essay is given in Appendix A of Hawking & Ellis (1973); see also Misner *et al.* (1973), p. 623. A proof of Birkhoff’s theorem appears in Appendix B of Hawking & Ellis (1973).

A very complete account of alternative theories of gravitation and the PPN formalism appears in Will (1993); see also Straumann (1991).

More details of gravitational lensing may be found in Chapter 4 of Peacock (1999) and in Hartle (2003), pp. 234–243 and more details of the radar echo data are given in Will (2001).

Problems

- 5.1 Laplace claimed that a star of the same density as the Earth with a diameter 250 times that of the Sun would be unable to shine. Verify this claim.
- 5.2 In the text it was claimed that the relative frequency of light from an atomic transition in the Sun (ν_1) and in a laboratory on Earth (ν_2) is approximately

$$\frac{\nu_1}{\nu_2} = 1 + \frac{GM_S}{c^2} \left(\frac{1}{R} - \frac{1}{R_S} \right).$$

Obtain a higher approximation for this ratio by taking into account the gravitational field of the Earth itself, and estimate the relative magnitude of this contribution.

- 5.3 Show that, according to Newtonian theory, for a planet of mass μ in a non-circular orbit around a Sun of mass M , the laws of conservation of energy (kinetic plus potential) and angular momentum $L = \mu a$ require that the equation for the orbit, $r(\phi)$, be (with $u = 1/r$)

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{a^2}.$$

- 5.4 Calculate the Einstein angle for (i) lensing within the galaxy of a star by an object of solar mass between us and the star, (ii) lensing of a source at a cosmological distance by a galaxy.
- 5.5 Prove Equation (5.97).
- 5.6 Prove Equation (5.98).
- 5.7 Find the radius at which light travels in a circular path round a body of mass M , (i) according to the General Relativity, (ii) according to Newtonian theory.

- 5.8 Show that the lines r variable (θ, ϕ, t constant), are geodesics of the Schwarzschild metric.
- 5.9 Consider two identical clocks A and B, synchronised and placed on the equator of the Earth. Clock A is taken to the North Pole and stays there for a year after which it returns to clock B on the equator. Ignoring the difference between the equatorial and polar radii of the Earth, use the Schwarzschild metric to calculate the resulting time discrepancy between A and B.
- 5.10 Recalculate the time discrepancy in the previous problem, taking into account the difference between the polar and equatorial radii of the Earth, given by the ‘flattening parameter’

$$\delta = \frac{R_E - R_P}{R_E} = 0.0034.$$

- 5.11 Using the symbols M, L and T for mass, length and time we may define *dimensions* of physical quantities, so that for example density ρ has dimensions $[\rho] = \text{ML}^{-3}$ – mass divided by volume. Check that the right and left hand sides of the field equation (5.24) have the same dimension.